

PatientTriage.ai

“Triage is a snapshot. Risk isn’t.”

A Continuous Safety Decision-Support Layer for Emergency Waiting Rooms • Accenture Innovation Challenge 2026

COMPETITION TRACK	CORE STACK	DEPLOYMENT	REPOSITORY
Round 2 Technical Prototype	Python 3.13 / FastAPI / React 19	Edge / On-Premise Air-Gapped	github.com/freya1705/PatientTriageAI

1. Project Overview & Introduction

PatientTriage.ai is an emergency department clinical decision-support system and continuous physiological safety surveillance layer. While traditional emergency triage treats patient prioritization as a static, one-time snapshot at hospital intake, PatientTriage.ai continuously tracks patient waiting times, vital sign trajectory velocity (ΔSpO_2 , ΔHR), data uncertainty, and active clinical attention. The intelligence layer performs continuous physiological trend analysis, uncertainty scoring, confidence decay, and dynamic attention-gap prioritization, while deterministic safety rules provide hard guardrails and clinicians retain final decision authority.

The platform answers the single most critical operational question facing emergency clinicians: “Who in the waiting room is no longer safe to keep waiting?”

2. The Core Problem & Competitive Moat (Vs. Legacy EHR Scores)

- 1. Silent Waiting Room Deterioration:** ESI Level 3/4 patients may worsen unmonitored; physiological decline may remain undetected until a subsequent reassessment or clinical deterioration becomes apparent.
- 2. Missing Vitals & Stale Data:** When vitals are missing, legacy systems default to low urgency. Under *Unknown is NOT Safe*, missing data heightens clinical vigilance.
- 3. The Attention Bottleneck:** Attended critical patients block static queues, while unattended deteriorating patients remain buried.
- Competitive Moat vs. Native EHRs (Epic EDI / Cerner MEWS):** Legacy EHR algorithms are designed for admitted inpatients in hospital beds and only score raw severity. PatientTriage.ai is purpose-built for the waiting room and uniquely factors in **physician coverage** and **evidence staleness decay**.

3. System Architecture: 3-Tier Layered Design

Tier	Responsibilities & Scope	Key Modules
Tier 1: Deterministic Safety Layer	Deterministic red-flags ($SpO_2 < 85\%$, $SBP < 75$ mmHg, FAST Stroke, pediatric stridor) that bypass statistical models. Counterfactual downgrade safety blocking.	safety_guardrails.py downgrade_guard.py
Tier 2: AI & Decision Support	Continuous vital trajectory velocity (ΔSpO_2 , ΔHR), dynamic confidence decay ($\tau_{staleness}$), uncertainty scoring ($Unknown \neq Safe$), Attention Gap re-ranking. Multimodal ingestion: BLE wearable oximetry rings/wristbands, kiosks, and nurse tablet walking rounds.	risk_engine.py deterioration_engine.py attention_gap_engine.py
Tier 3: Clinician Governance	Clinician override authority with mandatory justification recording, and immutable append-only audit logging.	audit_service.py OverrideModal.jsx

4. Core Intelligence Engines & Default Formula Weights

The Attention Gap Priority Equation:

Action Priority Score = ($w_r \times Risk + Urgency$) + ($w_d \times Deterioration$) + ($w_s \times Staleness$) + Wait Hazard + ($w_u \times Uncertainty$) - ($w_c \times Clinical Coverage$)

Default Parameter Bounds: $w_r = 1.0$ (base risk 0–100) • $w_d = +25$ to $+40$ pts ($\Delta SpO_2 \leq -5\%$ or $\Delta HR \geq +20$ bpm) • $w_s = +20$ to $+35$ pts (safety window expiry) • $w_u = +15$ to $+25$ pts (missing vitals / zero history) • $w_c = -35$ pts (when `is_attended = True`, surfacing unattended deteriorating patients to Rank #1).

5. Benchmark Cohort & Empirical Impact Evaluation

Evaluated across 20 synthetic clinical scenarios representing 5 systematic emergency department failure modes:

Performance Dimension	Traditional Static Triage	PatientTriage.ai	Benchmark Impact
Waiting Deterioration Catch Rate	0/20 detected	20/20 synthetic scenarios detected	100% Benchmark Coverage
Stale Observation Flagging	0/20 flagged	20/20 synthetic cases flagged (EXPIRED)	Zero Unmonitored Stale Waits
False Reassurance on Missing Vitals	High (Treated Normal)	0% False Reassurance (Unknown ≠ Safe)	Eliminates Under-Triage
Attention Gap Optimization	None (Attended Block)	Active (Elevates Unattended)	Optimized Clinician Utilization
Unsafe Priority Downgrades Blocked	0 Guardrails	100% Guarded (Proof Required)	100% Downgrade Guarded

**Note: Results reflect simulated evaluations across 20 synthetic clinical benchmark scenarios for prototype demonstration.*

6. Phased Implementation Roadmap & Precise Trial Endpoints

- **Phase 1 (Q3 2026):** Lab Validation • 20 synthetic scenarios verified across 33 automated tests; sub-15ms latency.
- **Phase 2 (Q4 2026):** Shadow Clinical Trial • Silent background deployment with Epic/Cerner via HL7 FHIR and CDS Hooks.
- **Phase 3 (Q1–Q2 2027):** Live Pilot • **Primary Safety Endpoint:** Reduction in Mean Time to Escalation (MTTE) (>45% faster); **Operational Endpoint:** Nurse false-alarm rate strictly < 2 alerts/nurse/shift; **Economic Endpoint:** ≥25% reduction in Left-Without-Being-Seen (LWBS).
- **Phase 4 (Q3 2027+):** Multi-Hospital Enterprise Network Scaling • Regional dashboarding and telemedicine integration.

7. Quick Start & Security Architecture

- # PowerShell 1-Command Launch: .\start.ps1 | Run 33 Automated Tests: python -m pytest -v
- **Zero PHI:** 100% synthetic physiological datasets • **Air-Gapped:** Zero third-party cloud LLM calls • **Audit Ledger:** Append-only event logs.

Regulatory & Safety Notice: PatientTriage.ai is a clinical decision-support research prototype developed for the Accenture Innovation Challenge 2026. All patient cohorts are synthetically generated. This system is not a certified medical device and does not replace professional clinical judgment.